UNDERGROWTH

Tide-aware autonomous drones for Willapa Bay oyster farms

FEATURES OVERVIEW
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Willapa Bay's sandbars, channels and burrowing-shrimp colonies change faster than any chart. Undergrowth puts small, quiet drones on the water so growers know where the ground is firm and safe **before** crews go out on the flats.

WORKING WITH THE TIDE CYCLE

Low tide	Rising tide	High tide
Beaches itself and waits. Cameras scan exposed beds for shrimp colonies.	Floats free. Uses the incoming water to lift, move and rotate oyster cages.	Sonar surveys channels, sandbars and depth to keep the chart current.

Map the bay

- Continuously update nautical charts: shifting sandbars, channels and depths
- Flag changing seabed and ground that is risky for aquaculture
- Monitor water temperature, currents, depth and other conditions
- Share maps and environmental data with NOAA and local fishermen and farmers

Tend the farm

- Lift and transport oyster cages on the incoming tide
- Move, position and rotate floating cages
- Track every cage's location and movement over time
- Inspect cages and spot obstacles with underwater cameras and sonar

Autonomous Exploration

- Operate around the clock without a human operator
- Wait for favorable tides instead of forcing immediate access
- Beach intentionally and resume moving when the tide returns
- Coordinate multiple drones peer-to-peer, with no central hub

Lighter footprint

- Articulating solar panels for power and wind-assisted propulsion
- Less fuel and fewer emissions than large diesel boats
- Quieter above and below the water; no unnecessary sonar pings
- Retrofit kits add sensors, navigation and autonomy to existing boats

Safer crews

Willapa Bay's tides move fast, channels shift without warning and shrimp-softened mud can trap a person. Drones scout the flats first, so crews only go out with a current map of safe ground.

Smarter planting

Shrimp colonies and sandbars keep moving. Tracking them over time lets growers predict where ground will stay firm and plant in the best locations.

Open to the community

Every grower can use the maps and app at no cost, and each connected farm improves the data for the whole bay.